# Back Translation Augmented Direct Preference Optimization for Neural Machine Translation


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***Abstract*—Neural machine translation systems can produce inaccurate translations even after supervised fine-tuning, motivating the use of preference-based post-training. However, the effectiveness of Direct Preference Optimization (DPO) partly depends on the quality of its preference pairs. This paper proposes a back-translation-based framework for constructing quality-controlled preference data for English-to-German translation. Using WMT14 parallel data, we treat each German sentence $s$ as the original text and its aligned expert-produced English translation $T(s)$ as the input to Gemma3-1B. The student translates $T(s)$ back into German, producing $\hat{T}(T(s))$, thereby learning the reverse translation direction of the expert translator. The original German sentence serves as the preferred response, while the generated back-translation is used as the rejected response only when its reference-based COMET22 score falls below an elbow-derived threshold of 0.72330. This process yields approximately 27,000 preference pairs from 100,000 candidates. We compare the baseline, standalone DPO, supervised fine-tuning (SFT), and sequential DPO+SFT. On the WMT14 test set, DPO improves five of six automatic evaluation metrics over the baseline, including COMET22 (0.7698 to 0.7810) and COMET-KIWI22 (0.7031 to 0.7476), although BLEU decreases (0.1572 to 0.1500). Sequential DPO+SFT achieves the best results on five metrics, including BLEU (0.1718), COMET22 (0.8119), and COMET-KIWI22 (0.7790), while standalone DPO achieves the best TER (0.7621). In blinded human evaluation, DPO receives a slightly higher preference than the baseline (22.2% versus 19.6%), while 58.2% of outputs are judged comparable. These findings suggest that combining reverse-direction preference learning with supervised fine-tuning is promising, although further controlled experiments are needed to isolate the contribution of COMET22-based filtering.**



***Index Terms*—Neural Machine Translation, Direct Preference Optimization, Back-Translation, Reinforcement Learning, Small Language Models**


## I. Introduction

Machine translation has evolved from rule-based systems relying on linguistic expertise and handcrafted grammars to statistical approaches and, more recently, neural methods based on deep learning. This progression has substantially improved translation capabilities, with neural machine translation (NMT) becoming the dominant paradigm through its use of large parallel corpora. Nevertheless, NMT systems can still omit content, mistranslate terminology, or produce unnatural and incomplete translations, even when trained on substantial parallel data. These limitations motivate post-training approaches that refine pretrained models beyond their initial supervised learning objectives. [1]

Preference-based optimization provides a way to refine models using comparative quality feedback in addition to standard supervised objectives. Direct Preference Optimization (DPO) is particularly attractive because it learns directly from preference pairs without a separately trained reward model, increasing the relative likelihood of preferred responses. [2] Although DPO has been widely studied for language-model alignment, its effectiveness depends in part on the quality and construction of preference pairs. Its benefits in machine translation, especially compared with conventional supervised fine-tuning (SFT), therefore remain an important empirical question.

We propose a back-translation-based framework for constructing quality-controlled preference pairs. Conventional back-translation is often used to create synthetic parallel examples for supervised training; here, the student-generated translation instead becomes the rejected side of a preference pair, while the trusted target-language text is preferred. The framework can use either monolingual data translated by an external expert (human or AI) or an existing parallel corpus, in which the human reference supplies the trusted target-side response. Our experiments use the latter setting with WMT14 English–German data: the student translates each English sentence into German, and its hypothesis is scored against the aligned reference with COMET22. Only hypotheses below the knee-point threshold are retained as rejected responses; the

reference remains the preferred response. [1]

We evaluate the framework with Gemma3-1B on English-to-German translation using approximately 27,000 filtered pairs. DPO improves five of six automatic metrics over the baseline, although BLEU decreases; in the direct ablation, SFT outperforms standalone DPO on five metrics, while DPO has the best TER. Sequential DPO+SFT achieves the best scores on five metrics. Human evaluation shows only a small preference difference between DPO and the baseline (22.2% versus 19.6%), with 58.2% of outputs judged comparable. The study addresses two related questions: whether DPO can improve a compact translation model when trained on reference-guided preference pairs, and how such pairs can be restricted to lower-scoring student hypotheses. Our contribution is the preference-data construction strategy and its evaluation in this setting, not a new DPO objective or a claim that DPO alone is superior to SFT. The experiments also do not isolate the causal effect of COMET22 filtering.

## II. Related Work

Related NMT research has explored synthetic-data generation, reinforcement learning (RL), and preference optimization. These approaches differ in whether they use synthetic parallel examples, explicit reward models, or pairwise comparisons to guide model updates.

Luu et al. [5] studied English-to-Basque translation using monolingual Basque data. They first translated Basque documents into English with a large model to create synthetic parallel data. After SFT, they generated Basque hypotheses from the English side and paired them with the original human-written Basque sentences as rejected and preferred responses, respectively. This pair structure is close to ours; our methodological addition is to screen each student-generated hypothesis with COMET22 before admitting it as a rejected response. Their preliminary experiments compare five SFT/DPO allocations using 20,000 examples and find that SFT-only performs strongest overall, while adding DPO yields small or inconsistent changes. Their rejected hypotheses, generated by NLLB-3.3B, are not COMET-screened at the preference-pair construction stage. This does not mean their pipeline has no filtering: they separately clean the synthetic training corpus and use quality estimation to select some evaluation documents. The relevant difference is the absence of COMET screening of rejected hypotheses. Their language pair, data setting, and model also differ from ours; our own filtering decisions have not been independently validated by human annotators.

Zhang et al. [6] used in-domain source-side monolingual data and a ranking-based reward model trained on a small in-domain parallel set to guide RL fine-tuning of a pre-trained NMT model. Their approach demonstrates the value of monolingual data and quality feedback for domain adaptation. Unlike our method, it relies on a learned reward model, whose judgments may influence the optimization, rather than direct preference optimization. The reward-model stage also introduces an additional component that must be trained before it can guide the translation model.

Uhlig et al. [7] proposed Direct Quality Optimization (DQO), a DPO variant that uses a pretrained translation-quality-estimation model as a proxy for human preference in an online optimization procedure. By contrast, our COMET22 score is used offline to select rejected hypotheses during data construction, not as an online reward or replacement for the DPO objective. Thus, the metric determines which examples enter the training set, while the DPO loss remains the optimization objective. DQO reports gains across multilingual models in automatic metrics and human evaluation.

Vajda et al. [8] constructed synthetic English-to-Slovene preference pairs by comparing translations generated by different LLMs, using heuristics to identify obvious errors and metrics such as COMET to distinguish output quality. Their approach compares independently generated hypotheses and selects the preferred output from those candidates. Our setting instead fixes the aligned human reference as preferred and uses COMET22 to screen one student-generated hypothesis for the rejected response; it does not rank two model outputs. Vajda et al. report COMET gains and greater consistency in avoiding language and formatting errors relative to their baseline generators.

Overall, prior work differs in how it creates preference signals: Luu et al. use human references and generated hypotheses without COMET screening at the pair-construction stage; Vajda et al. compare outputs from two LLMs; and DQO uses quality estimation during optimization. Our specific addition is an offline COMET22 threshold applied to the student hypothesis before DPO pair construction. The metric is used to decide whether the student hypothesis enters the preference dataset, not to choose between two model-generated candidates, revise the human reference, or provide an online reward during optimization. Although the resulting pipeline improves five of six automatic metrics over our baseline, the absence of a matched unfiltered-DPO experiment and an independent human validation of the filtering decisions means that these gains cannot be attributed specifically to COMET22 screening.

Shen et al. [9] proposed post-editing-guided reinforcement learning (PEGRL), a two-stage framework that treats post-editing as an auxiliary task. Post-edited outputs are used to stabilize training and provide additional optimization signals for the primary translation objective.

## III. Methodology

Our framework constructs quality-controlled preference pairs and uses them to fine-tune a baseline language model, or *student model* $\pi_\theta$. It contrasts a trusted target-side translation with a student output estimated to have lower quality. Figure 1 summarizes corpus preparation, candidate generation, filtering, and DPO fine-tuning.

[1] All code, models, trained checkpoints, and experimental resources associated with this work are publicly available at github.com/mehrdadghassabi/Amestris to support reproducibility and further research.

The preference dataset is $\mathcal{D} = \{(x, y_w, y_l)\}$, where $x$ is the translation instruction and input, $y_w$ is the preferred response, and $y_l$ is the rejected response. The preferred response is an aligned human translation; the rejected response is generated by the student and retained only when its COMET22 score against the source and reference is sufficiently low. DPO is trained on both responses in each pair, contrasting the trusted reference with the lower-scoring hypothesis. Thus, the method does not discard the low-scoring translation from training: it deliberately uses that translation as the rejected side of the preference pair. The source sentence and translation instruction are shared by both responses, allowing DPO to compare the two candidate translations under the same input.

In the general setting, a monolingual corpus is segmented into sentences, and each sentence $s$ is translated by an expert translator $T(\cdot)$. The student $\pi_\theta$ then translates $T(s)$ back into the original language, producing $\hat{T}(T(s))$. The expert translator may be human or AI-based. An existing parallel corpus supplies the same information without a separate expert-translation step. In our WMT14 experiment, the German reference is $s$ and its aligned English sentence is $T(s)$; the student therefore translates English into German, the same direction used in evaluation (Section IV). This distinction resolves the direction ambiguity: the German sentence is the preferred output, not the model input.

For each candidate, we compute the reference-based score $q(s) = \text{COMET22}(T(s), \hat{T}(T(s)), s)$ using the English source, the student's German hypothesis, and the aligned German reference. We retain the candidate only when $q(s) < \tau$, where $\tau$ is the knee point of the score distribution. The resulting preference triplet is

$$x = \text{prompt} + T(s),$$
$$y_w = s, \qquad y_l = \hat{T}(T(s)), \qquad q(s) < \tau.$$

Here, $x$ contains the instruction to translate $T(s)$ into German, $y_w = s$ is the German reference, and $y_l = \hat{T}(T(s))$ is the student's German hypothesis. Candidates with $q(s) \geq \tau$ are excluded because they are not part of the lower-scoring group targeted for rejected responses. In this WMT14 experiment, COMET22 is the sole filtering criterion; BLEU-based filtering is not applied.

Table I shows an actual WMT14 preference pair: the English source is the input, the aligned German reference is preferred, and the student's German hypothesis is rejected. The hypothesis retains an English phrase, mistranslates "my Group" and "object to that," and contains grammatical errors.

The COMET22 threshold applies only to the student hypothesis, not to the human reference. We identify the score-distribution knee using the Kneedle/elbow method [12] and apply it during preference-data construction, before DPO training. The purpose is to avoid labeling relatively high-scoring student translations as rejected examples and to focus preference learning on the lower-scoring region. Because COMET22 is an automatic quality estimator, a score below the threshold indicates lower estimated quality, not proof that every retained translation is incorrect.

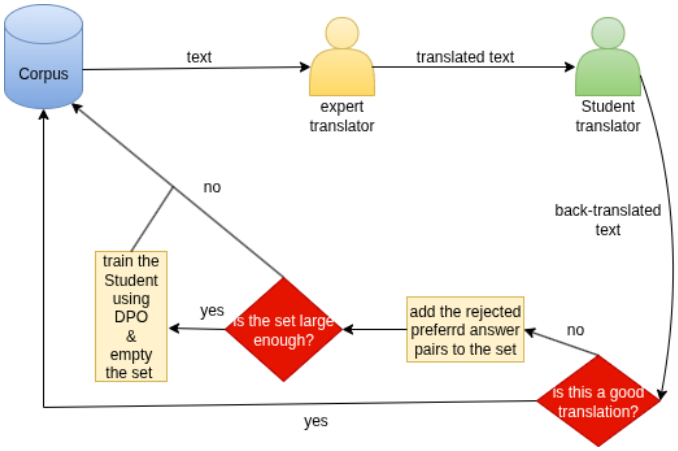



Fig. 1. Overview of the proposed method

We fine-tune $\pi_\theta$ on $\mathcal{D}$ using DPO with LoRA [11]. DPO learns directly from pairwise preferences without a separate reward model, while LoRA updates low-rank matrices to reduce compute and memory costs. The loss for one triplet is

$$\mathcal{L}_{\text{DPO}}(\pi_\theta) = -\log \sigma \left( \beta \left[ \log \frac{\pi_\theta(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \log \frac{\pi_\theta(y_l|x)}{\pi_{\text{ref}}(y_l|x)} \right] \right),$$

where $\pi_{\text{ref}}$ is a fixed reference model (typically a frozen copy of the pretrained student), $\beta$ is a temperature parameter controlling optimization sharpness, and $\sigma(\cdot)$ is the sigmoid function. The objective increases the preferred response's likelihood relative to the rejected response, measured against the fixed reference model. In this way, the model is optimized using the difference between its relative likelihoods for the two responses rather than by fitting a separately learned scalar reward.

In the full framework, the preference dataset is refreshed after each student update. Hardware constraints limited the experiments reported here to one iteration; further iterations are left for future work.

## IV. Evaluation

We evaluated English-to-German translation with automatic metrics and pairwise human assessment, using Gemma3-1B as the baseline. Hardware constraints limited the proposed framework to one training iteration.

### A. Experimental Setup

We used the WMT14 English–German parallel dataset [13]. Because each English sentence is paired with a human German translation, no separate expert-translation step was required. We randomly sampled approximately 100,000 aligned pairs from the WMT14 training partition. The English sentence was the student input, the German reference was preferred, and the student's German hypothesis was the rejected response, subject to the filtering rule below. This setup uses the existing reference translation as the preferred answer rather than asking an additional translator to produce one.

TABLE I
ILLUSTRATIVE ENGLISH-TO-GERMAN PREFERENCE PAIR.

| **Field** | **Example** |
|---|---|
| Language model input ($x$) | Translate the following English sentence into German.<br>Madam President, if the vote records correctly how my Group voted I shall not, and cannot, object to that. |
| Preferred response ($y_w$) | Frau Präsidentin, wenn aus der Abstimmung einwandfrei hervorgeht, wie meine Fraktion abgestimmt hat, dann werde und kann ich nichts dagegen sagen. |
| Rejected response ($y_l$) | Madam Präsident, wenn die Wahlprotokolle korrekt sind, wie ich meine Gruppe gewählt hat, werde ich nicht, und kann nicht, gegen diese objektivieren. |
| Rejected-response scores | COMET: 0.4782; BLEU: 1.9968. |

The official WMT14 test partition (approximately 3,000 sentence pairs) was reserved for final evaluation and was not used for preference-pair generation, filtering, or fine-tuning. Training and test examples therefore came from disjoint partitions.

Gemma3-1B was selected as the baseline because its compact size made inference and fine-tuning feasible with our available hardware and accessible cloud resources, including free cloud-computing tiers. Its relatively small parameter count made it practical for the experiments while providing a baseline for measuring the effects of post-training. The conclusions should nevertheless be interpreted within this model scale rather than generalized to larger translation systems.

From the 100,000 candidates, we retained hypotheses with COMET22 scores below 0.72330, the elbow threshold, and did not apply BLEU-based filtering. The threshold was applied to the student hypothesis, not the human reference. Filtering excluded approximately 73,000 candidates (73%) and retained approximately 27,000 pairs (27%) for LoRA-based DPO fine-tuning. Each retained pair therefore contains both the human reference and the low-scoring hypothesis: the latter remains in training as the rejected response. The reduction in dataset size does not by itself establish a downstream benefit from filtering. The resulting model is referred to as *amestris-1b-dpo*.[2] Training took approximately eight hours on two Kaggle T4 GPUs. Table II lists the hyperparameters, and Figure 2 shows the score distribution.

TABLE II
TRAINING HYPERPARAMETERS.

| | |
|---|---|
| Number of epochs | 1 |
| Warmup ratio | 0.03 |
| $\beta$ (DPO temperature parameter) | 0.1 |
| Gradient accumulation steps | 4 |
| LoRA rank | 32 |
| LoRA alpha | 32 |
| LoRA dropout | 0.05 |
| Target modules | All linear layers |
| BLEU filtering | Not applied |
| COMET22 threshold | 0.72330 |
| Preference pairs retained | 27,000 |

[2]Available at huggingface.co/gaokerena/amestris-1b-dpo

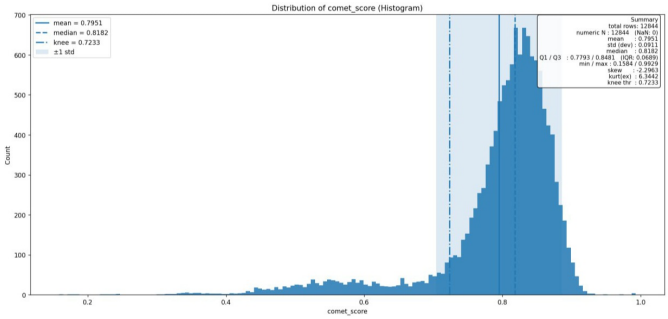



Fig. 2. COMET22 scores of student-generated translation hypotheses used for preference-pair construction. The knee threshold (0.72330) retains lower-scoring hypotheses as rejected responses and excludes higher-scoring candidates.

### B. Automatic Evaluation

We evaluated *gemma3-1b* and the DPO-fine-tuned model, *amestris-1b-DPO*, on the held-out WMT14 test partition using six automatic metrics (Table III).

Higher scores are better for BLEU [10], COMET22, COMET_KIWI22, METEOR [16], and chrF++ [18]; lower TER [17] is better. BLEU measures n-gram overlap between the system output and reference. COMET22 is a reference-based neural metric trained on WMT22 human judgments [15] and evaluates the source, output, and reference, whereas COMET_KIWI22 estimates quality from the source and output without requiring a reference. METEOR combines lexical precision and recall with stemming and synonym matching; TER measures the edit operations needed to transform an output into the reference; and chrF++ computes an F-score over character n-grams, capturing some morphological and orthographic variation.

TABLE III
AUTOMATIC EVALUATION RESULTS ON THE WMT14 TEST SET.

| | Gemma3-1B (baseline) | Amestris-1B-DPO |
|---|---|---|
| BLEU ↑ | **0.1572** | 0.1500 |
| COMET22 ↑ | 0.7698 | **0.7810** |
| COMET_KIWI22 ↑ | 0.7031 | **0.7476** |
| METEOR ↑ | 0.3861 | **0.3969** |
| TER ↓ | 0.7765 | **0.7621** |
| chrF++ ↑ | 0.4193 | **0.4382** |

DPO improves five metrics over the baseline: COMET22 (+0.0112), COMET_KIWI22 (+0.0445), METEOR (+0.0108),

chrF++ (+0.0189), and TER (a decrease of 0.0144). BLEU is lower (0.1500 versus 0.1572), so the gains are not uniform. These changes indicate improvements in several semantic, lexical, and edit-based measures, but not in n-gram overlap. In particular, the reference-free COMET_KIWI22 increase suggests a gain under a metric that does not use the reference at scoring time, while the BLEU decrease highlights a trade-off across evaluation criteria. Because this checkpoint was trained on the 27,000 COMET22-filtered pairs and no matched unfiltered-DPO condition was evaluated, these results measure the complete filtered-data pipeline but do not isolate the effect of filtering. The BLEU decrease likewise cannot be attributed to the absence of BLEU-based filtering.

Figure 2 shows a smaller lower-scoring group and a larger higher-scoring concentration. The threshold (0.72330) retains the former as rejected responses and excludes the latter, aiming to avoid treating relatively high-scoring hypotheses as negative examples. The histogram is used to motivate the knee-point selection, but its shape alone does not establish that the retained and excluded groups are uniformly erroneous and correct, respectively. The histogram and COMET22 scores are automatic estimates, not proof that every retained candidate is erroneous or every excluded candidate is correct. The filtering decisions were not independently validated by human annotation (Section VI).

### C. Human Evaluation

We conducted pairwise human evaluation of the baseline *gemma3-1b* and *amestris-1b-dpo* on held-out WMT14 instances. One evaluator, an English–German expert with the language expertise required to assess the source and target texts, assessed each English source and its two German translations. Outputs were anonymized, and model identities were concealed; the evaluator was not told which system produced either candidate.

The evaluator considered semantic adequacy with respect to the English source, completeness, grammatical correctness, fluency, and lexical appropriateness, and assigned one of three judgments: first output better, second output better, or comparable. Each pair was judged as a whole rather than by assigning separate numerical scores to each criterion. After annotation, the anonymized outputs were mapped back to their systems to calculate preference rates. Evaluation examples and annotations are public.[3] Example references below use the released data's row numbers.

As shown in Figure 3, the evaluator preferred DPO in 22.2% of cases and the baseline in 19.6%; 58.2% were judged comparable. The 2.6-percentage-point difference indicates only a modest preference for DPO. The large comparable category also suggests that many outputs were judged similar in overall quality rather than clearly favoring either system. Given the single-evaluator protocol, these percentages are descriptive and do not establish a generalizable human preference advantage.

Selected examples suggest occasional gains in completeness: in row 30, the baseline leaves parts of the source untranslated or mixes English and German, whereas the DPO output provides a more complete German rendering and more closely follows the reference sentence's overall structure.

Rows 34 and 37 show occasional fluency improvements, including more natural word order, smoother clause structures, and more idiomatic lexical choices. These are observations from individual examples and do not indicate that fluency improved uniformly across the test set.

The evaluator's annotations also identify examples of improved lexical selection and domain-specific terminology. Row 10 illustrates a more appropriate lexical choice, but also shows that better terminology does not guarantee numerical fidelity. In row 24, the DPO output was judged closer to the reference in preserving contextual relationships, although it retained some imperfections.

The human evaluation also identified errors. Several outputs contained hallucinated names, dates, or numerical values. Other examples showed corruption of personal names, geographical locations, and institutional titles; row 2973 illustrates that named entities may remain incorrect even when the surrounding translation is fluent. In some cases, the model produced a fluent German sentence corresponding to an adjacent source sentence rather than the current input, substantially reducing adequacy. A small number of outputs also contained decoding failures, malformed strings, or text not aligned with the input.

Overall, the examples show possible improvements in completeness, fluency, and lexical choice, alongside factual, named-entity, and sentence-alignment errors. The examples therefore illustrate both potential strengths and remaining failure modes; they should not be interpreted as a systematic error analysis or as evidence that the model improves every aspect of translation. Together with the modest preference difference and single-evaluator design, these findings qualify the automatic results rather than establish a consistent human-evaluated advantage for DPO.

## V. Ablation Study

To examine the relative contributions of preference-based training and supervised fine-tuning, we compared the baseline, standalone DPO, standalone SFT, and sequential DPO followed by SFT (DPO+SFT). [4] The configurations were assessed using the same automatic metrics and, as described in the experimental setup, the same underlying dataset setting.

Luu et al. [5] provide an external reference for DPO without COMET screening of rejected hypotheses. Their preliminary experiments (Section 4.2, Table 3) compare five SFT/DPO allocations totaling 20,000 examples: 20k/0, 15k/5k, 10k/10k, 5k/15k, and 0/20k. SFT-only (20k/0) performs strongest overall on FLORES-200 dev and NTREX. With 15k SFT and 5k DPO, results are mixed: on FLORES-200 dev, COMET22 increases from 79.41 to 79.54 and chrF++ from 42.93 to 43.20, while BLEU falls from 10.61 to 10.53 and BLEURT

[3] github.com/Mehrdadghassabi/Amestris/assets

[4] available at huggingface.co/gaokerena/amestris-1b-dpo-sft and huggingface.co/gaokerena/amestris-1b-sft

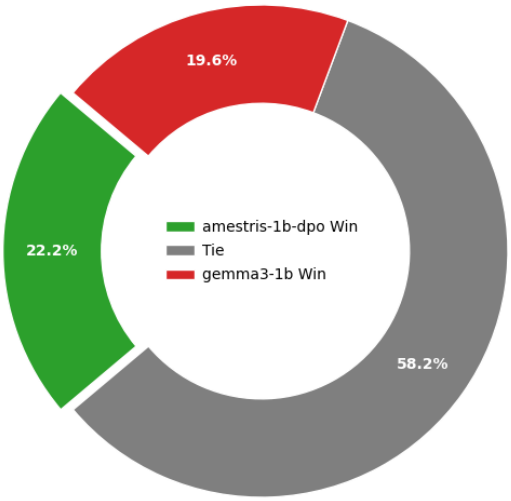



Fig. 3. Pairwise human-evaluation preferences for Amestris-1B-DPO and Gemma3-1B.

from 67.22 to 66.89. On NTREX, chrF and chrF++ rise slightly (46.34 to 46.42 and 41.09 to 41.11), while BLEU, COMET22, COMETKIWI22, and BLEURT decline. Their DPO-only setting is below SFT-only on all six reported metrics on both sets. This suggests that DPO over generated hypotheses may provide a weak or noisy signal in that setting, but does not show that COMET screening would solve the problem. Importantly, Luu et al. do not screen rejected hypotheses with COMET at pair-construction time, although they separately clean their synthetic training corpus and use quality estimation to select some evaluation documents. Their language pair, model, and data regime also differ from ours. Vajda et al. [8], by contrast, report COMET improvements from DPO for English-to-Slovene, illustrating that results vary by task and preference construction. In our English-to-German experiment, filtered DPO improves five of six metrics over the baseline; however, without a matched unfiltered condition, we cannot attribute those gains to COMET22 filtering. SFT maximizes the likelihood of reference translations, whereas DPO additionally learns from rejected responses.

Table IV shows that SFT outperforms standalone DPO on five metrics (BLEU, COMET22, COMET_KIWI22, METEOR, and chrF++); DPO has the best TER. DPO+SFT achieves the best scores on the other five metrics, while DPO retains the best TER. Compared with SFT alone, sequential DPO+SFT improves five metrics but has a slightly higher (worse) TER. Thus, DPO's gains over the baseline do not mean it outperforms SFT. The results suggest that DPO may provide a complementary signal when followed by SFT in this setup, but no configuration is best on every metric, and the ranking may differ on other datasets or models.

COMET22 filtering was intended to pair human references with lower-scoring student hypotheses and thereby provide a more focused negative signal. It reduced approximately 100,000 candidates to 27,000 pairs, but this reduction is only a data-selection result and does not establish that the retained data improve DPO. We neither trained a matched DPO model on the full unfiltered candidate set nor independently validated the filtering decisions with human assessment; the present ablation therefore does not isolate filtering's contribution.

Luu et al.'s results are relevant external evidence, not a substitute for this missing ablation. For the 15k-SFT/5k-DPO allocation, COMET22 changes from 79.41 to 79.54 on FLORES-200 dev but from 77.55 to 77.32 on NTREX, while BLEU decreases on both sets; their DPO-only setting is also weaker than SFT-only. The findings are useful as context because they show that adding DPO to an SFT system does not guarantee improvements when rejected hypotheses are not screened in the same way. However, the comparison remains external: language pair, model, data, and experimental settings differ. These findings motivate testing whether lower-quality rejected hypotheses provide a more informative preference signal. Because Luu et al. do not report a matched COMET-filtered condition, their results cannot quantify the effect of filtering. That requires otherwise identical DPO models trained on the same candidate pool with and without COMET screening.

TABLE IV
COMPARISON OF BASELINE TRAINING WITH DPO AND SFT

| | Gemma3-1b (baseline) | Amestris -1b-DPO | Amestris -1b-SFT | Amestris-1b -DPO-SFT |
|---|---|---|---|---|
| BLEU ↑ | 0.1572 | 0.1500 | 0.1701 | **0.1718** |
| COMET22 ↑ | 0.7698 | 0.7810 | 0.8007 | **0.8119** |
| COMET KIWI22 ↑ | 0.7031 | 0.7476 | 0.7738 | **0.7790** |
| METEOR ↑ | 0.3861 | 0.3969 | 0.4297 | **0.4325** |
| TER ↓ | 0.7765 | **0.7621** | 0.7840 | 0.7847 |
| chrF++ ↑ | 0.4193 | 0.4382 | 0.4487 | **0.4524** |

## VI. FUTURE WORK

The findings are mixed: filtered DPO improves five metrics over the baseline, but SFT outperforms standalone DPO on five metrics, and DPO+SFT leads on five. Human evaluation shows only a 2.6-percentage-point preference difference, with 58.2% of outputs judged comparable. The results therefore do not establish standalone DPO as the best strategy, isolate the effect of COMET22 filtering, or demonstrate a substantial human preference advantage.

First, the framework should be tested in specialized domains, including medical, legal, and technical translation, to assess its ability to improve domain-specific terminology, preserve specialized meaning, and reduce translation errors. These settings would also help determine whether COMET22-based selection behaves consistently when the vocabulary and error patterns differ from those in general-domain WMT14 data.

Second, the filtering stage needs independent validation and threshold-sensitivity analysis. The 27,000 training pairs were selected automatically with the COMET22 elbow threshold; we did not obtain human judgments for retained and discarded

candidates, compare those judgments with COMET22, or retrain using alternative thresholds or selection rules. The human evaluation reported here assesses final model outputs, not the filtering decisions, and therefore cannot be used as independent evidence that the filtering labels agree with human assessment. We also did not compare matched DPO models trained on filtered and unfiltered versions of the same candidate pool. Thus, the observed gains should be attributed to the complete DPO pipeline, not specifically to COMET22 filtering. Future experiments should (i) have qualified annotators assess representative retained and discarded pairs, (ii) compare standard DPO on unfiltered pairs with COMET-filtered DPO, SFT-only, and the pretrained baseline under matched data and training budgets, and (iii) compare the elbow threshold with fixed COMET cutoffs, quantile-based thresholds, and size-matched random selection. Results should be reported for COMET and BLEU, as well as the other metrics, to reveal threshold sensitivity and metric trade-offs.

Third, human evaluation should involve multiple independent bilingual evaluators. Although our evaluator was an English–German expert and model identities were concealed, a single rater cannot provide inter-annotator agreement or establish that the observed preferences are stable across annotators. Future studies should also randomize output order, use a consistent evaluation interface, and report agreement alongside preference rates.

Finally, future work should test alternative pair-sampling strategies, preference objectives such as SimPO [19] and ORPO [20], and parameter-efficient methods such as QLoRA [21]. The contributions of back-translation, DPO, and SFT should also be examined across more language pairs, datasets, and evaluation protocols.

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